

Question 15

1 mark

Given the polynomial function $P(x) = x^4 - 5x^2 - 2x + 6$, if $P(1) = 0$, identify which statement is true.

- a) The y -intercept is 1.
- b) $P(x)$ has a factor of $(x + 1)$.
- c) The graph has a zero at 1.
- d) The graph has a zero at -1 .

Question 16

1 mark

There are 6 different books that are being distributed evenly amongst three people. Identify which expression represents the number of possible combinations.

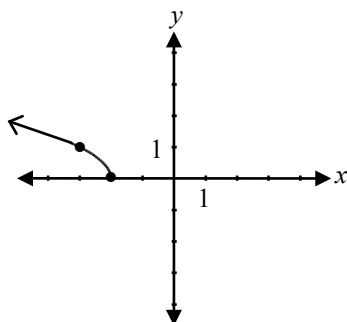
- a) ${}_6C_2 \cdot {}_6C_2 \cdot {}_6C_2$
- b) ${}_6C_2 \cdot {}_4C_2 \cdot {}_2C_2$
- c) ${}_2C_2 \cdot {}_2C_2 \cdot {}_2C_2$
- d) $3 \cdot {}_6C_2$

Question 17

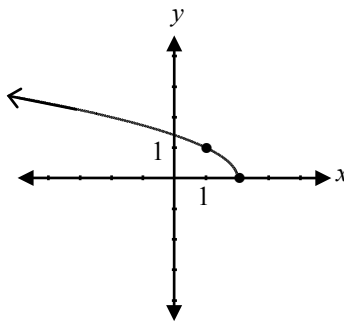
1 mark

Identify the graph that corresponds to the function $f(x) = -\sqrt{(x-2)}$.

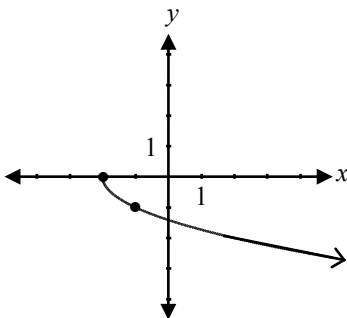
a)



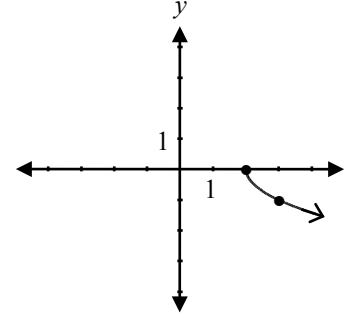
b)



c)



d)



Question 18

1 mark

Solve:

$$7^{\log_7 2} = x$$

a) $x = 1$

b) $x = 2$

c) $x = 7$

d) $x = 49$

Question 19

1 mark

Identify the equation that has a general solution of $\left. \begin{array}{l} \theta = \frac{\pi}{6} + 2\pi k \\ \theta = \frac{5\pi}{6} + 2\pi k \end{array} \right\}$ where $k \in \mathbb{Z}$.

a) $\sin \theta = \frac{1}{2}$

b) $\cos \theta = \frac{1}{2}$

c) $\sin \theta = \frac{\sqrt{3}}{2}$

d) $\cos \theta = \frac{\sqrt{3}}{2}$

Question 20

1 mark

Identify the function that has a domain of $x \leq -2$ and a range of $y \geq 3$.

a) $y = \sqrt{x+2} + 3$

b) $y = \sqrt{-(x+2)} + 3$

c) $y = -\sqrt{x-2} - 3$

d) $y = -\sqrt{-(x-2)} - 3$

Question 21

1 mark

Given $f(x) = 3x + 2$, identify $f^{-1}(x)$.

a) $f^{-1}(x) = -3x - 2$

b) $f^{-1}(x) = 2x + 3$

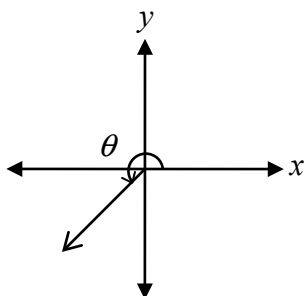
c) $f^{-1}(x) = \frac{x}{3} - 2$

d) $f^{-1}(x) = \frac{x-2}{3}$

Question 22

1 mark

Identify a possible value for the angle θ sketched in standard position.



a) 2

b) 3

c) 4

d) 5

Question 23

3 marks

116

Solve the following equation:

$$\log_3(x+3) + \log_3(x-5) = 2$$

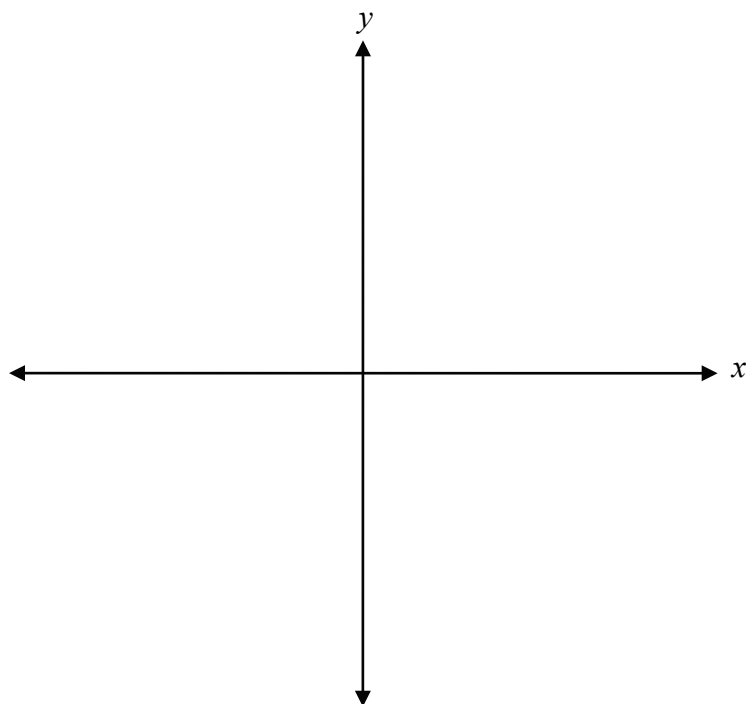
Question 24

1 mark

117

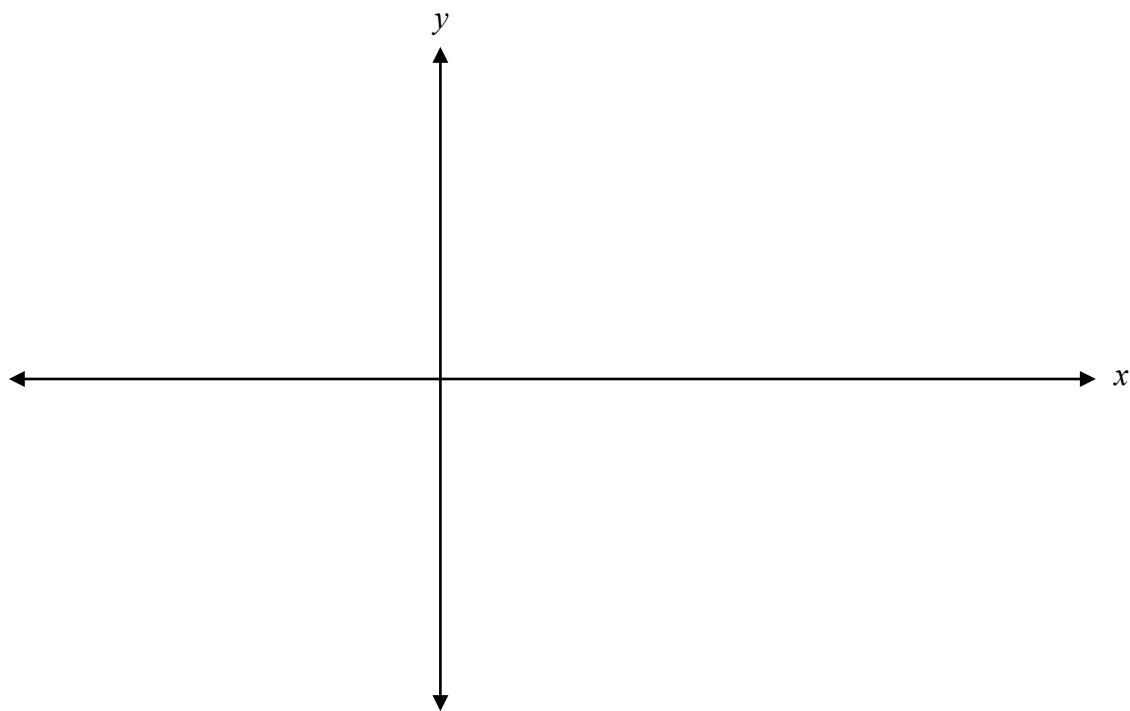
State a coterminal angle for $\theta = \frac{9\pi}{4}$.

Sketch the graph of the function $f(x) = \frac{2x+2}{x^2-1}$.



Justify why the binomial expansion of $(x + x^3)^7$ does not have a term containing x^{10} .

Sketch the graph of $y = -\sin\left(\frac{\pi}{2}(x-1)\right) + 3$ over the domain $[0, 6]$.



Question 28

2 marks

121

When $P(x) = 3x^4 - kx^3 + 5x - 14$ is divided by $(x + 2)$, the remainder is -8 .

Determine the value of k .

Question 29

a) 3 marks b) 1 mark

122
123

Given that $\cos \alpha = \frac{7}{12}$ where α is in quadrant IV, and $\sin \beta = \frac{3}{5}$ where β is in quadrant I, determine the exact value of:

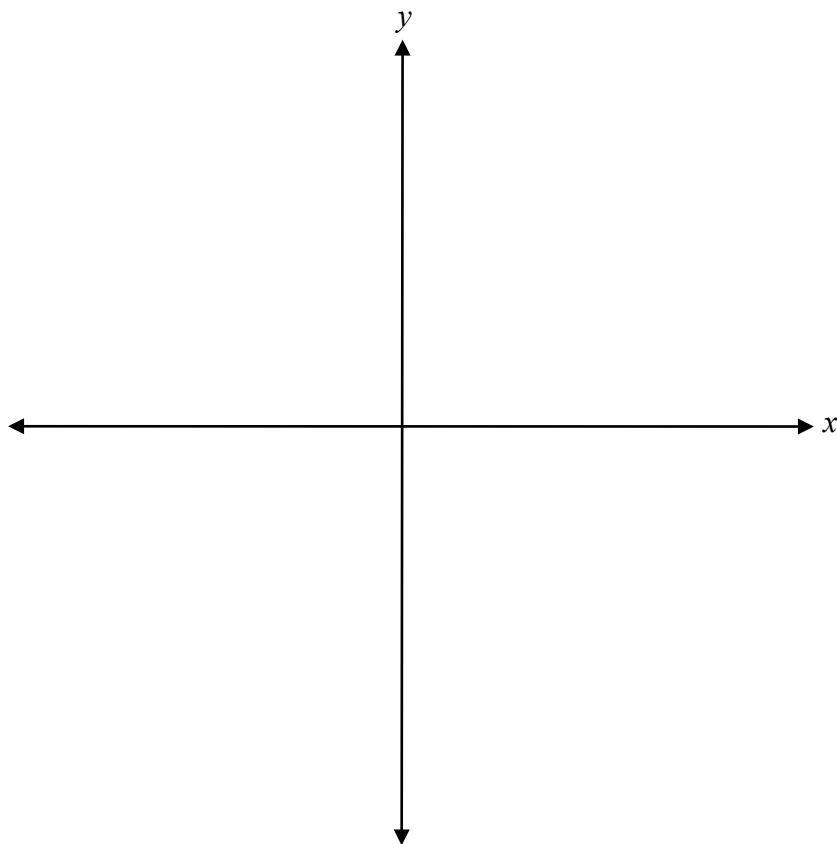
a) $\sin(\alpha - \beta)$

b) $\csc(\alpha - \beta)$

Describe the difference between the graph of $f(x) = \frac{7(x+2)}{x+2}$ and the graph of

$$g(x) = \frac{7(x-2)}{x+2} \text{ at } x = -2.$$

Sketch the graph of $f(x) = 3^x + 2$.



Question 32

3 marks

126

Solve algebraically:

$${}_nC_3 = n - 2$$

Describe the error that was made when solving the following equation:

$$\sin^2 \theta + \sin \theta - 2 = 1$$

$$\sin^2 \theta + \sin \theta = 3$$

$$\sin \theta (\sin \theta + 1) = 3$$

$$\sin \theta = 3 \quad \sin \theta + 1 = 3$$

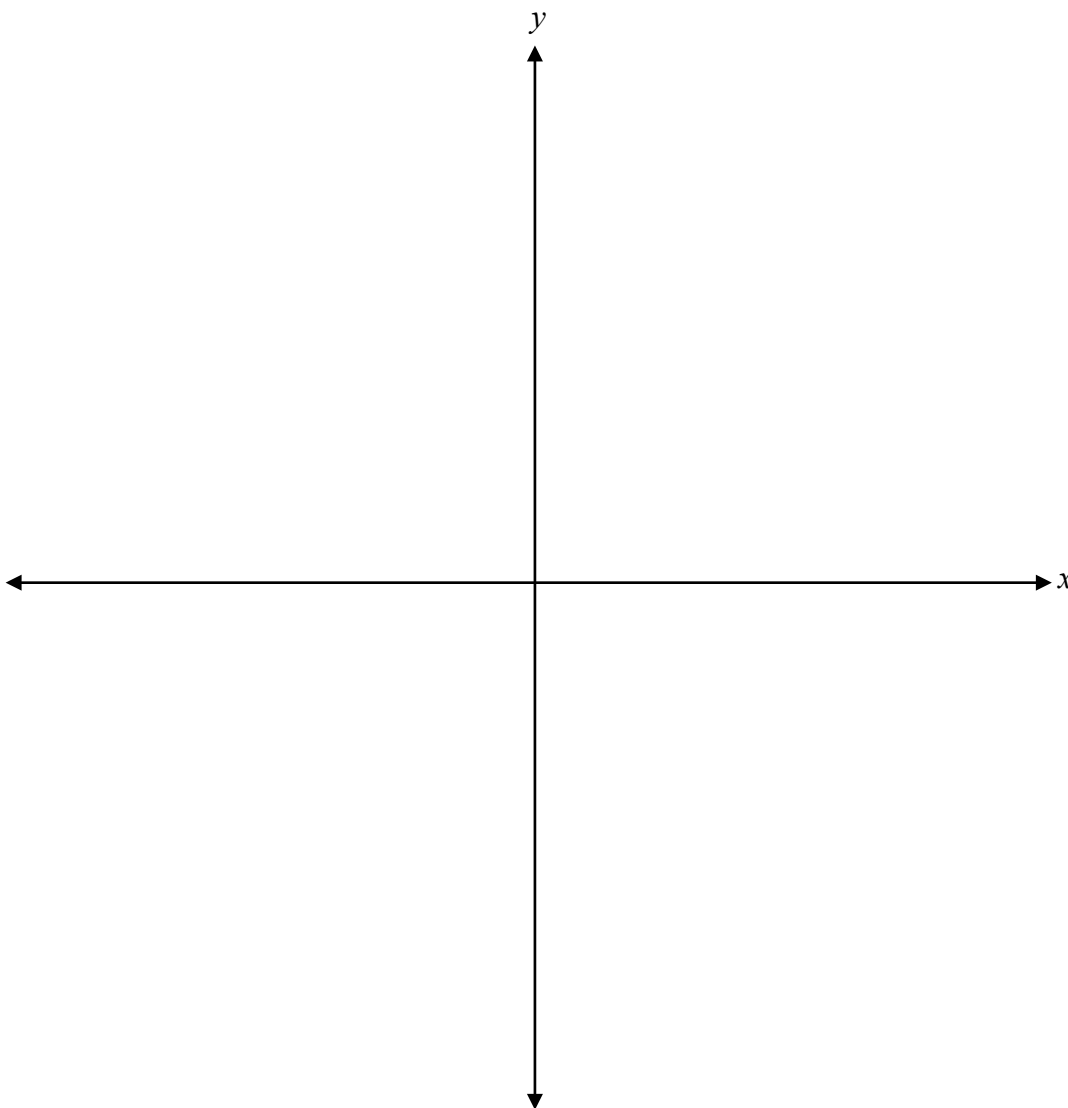
$$\sin \theta = 2$$

$\therefore$ No solution

$\therefore$ No solution

Sketch the graph of the polynomial function with the following characteristics.

- a y -intercept of -9
- zeroes at -1 and 3
- the zero at -1 has a multiplicity of 1 and the zero at 3 has a multiplicity of 2



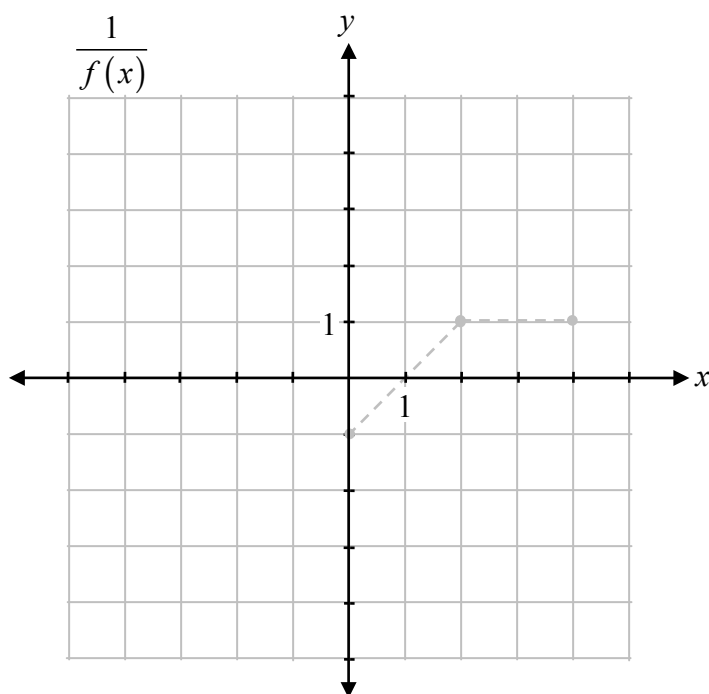
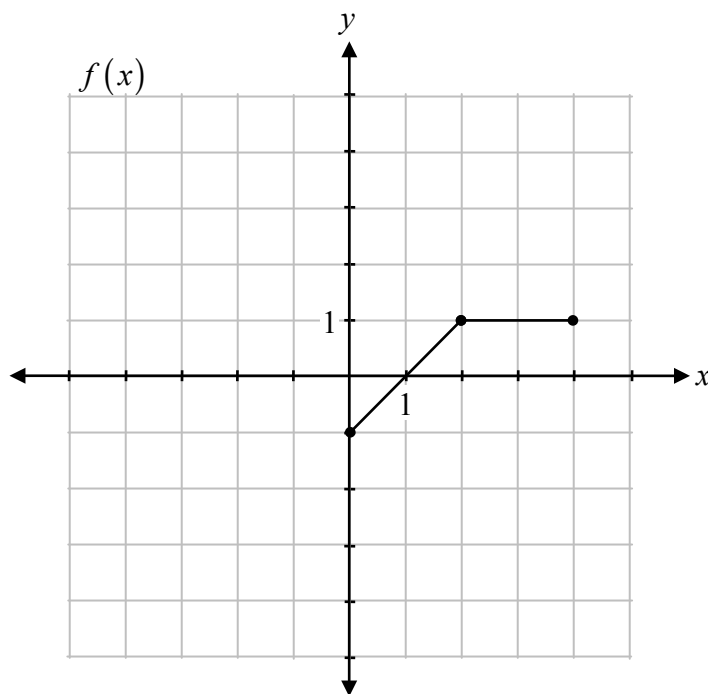
Question 35

2 marks

129

Given $\cot \theta = -\frac{1}{3}$, where θ is in quadrant II, determine the exact value of $\sin \theta$.

Given the function $f(x)$, sketch the graph of the reciprocal, $\frac{1}{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

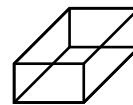
No marks will be awarded for the graph of $f(x)$.

Question 37

3 marks

131

The volume of a planter, in the shape of a rectangular prism, can be modelled by the polynomial function $V(x) = x^3 + 3x^2 - 34x + 48$.



Determine the factors of the function, $V(x)$, which represent possible dimensions of this planter.

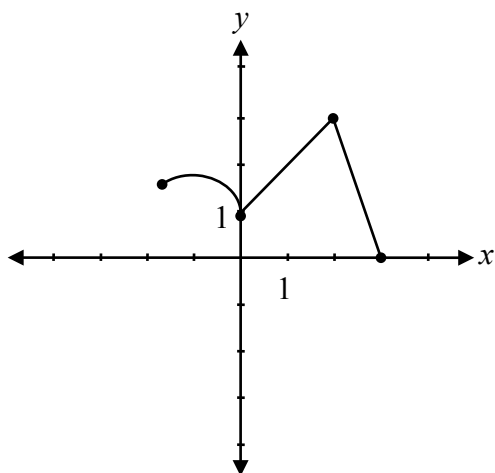
$V(x) =$ _____

Question 38

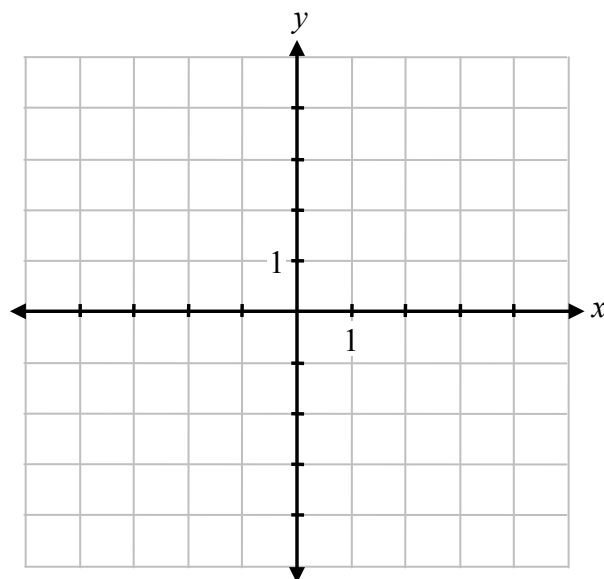
1 mark

132

Describe how to determine the range of the inverse of the following graph.



Sketch the graph of the function $y = \sqrt{2x} + 1$.



Question 40

a) 2 marks b) 1 mark

134
135

Given the following characteristics of a sinusoidal function:

- an amplitude of 2
- a vertical translation down 3 units
- a period of $\frac{\pi}{4}$

a) Determine an equation of this sinusoidal function in the form $y = a \sin b(x - c) + d$.

b) Determine the range of this function.

Range:

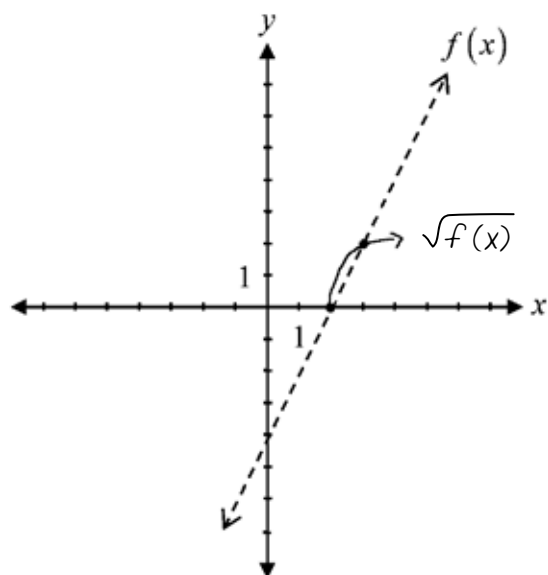
Question 41

1 mark

136

Suah was given the graph of $f(x)$ and asked to graph $y = \sqrt{f(x)}$.

Her solution is given on the graph below.



Describe the error Suah made when sketching the graph of $y = \sqrt{f(x)}$.

Question 42

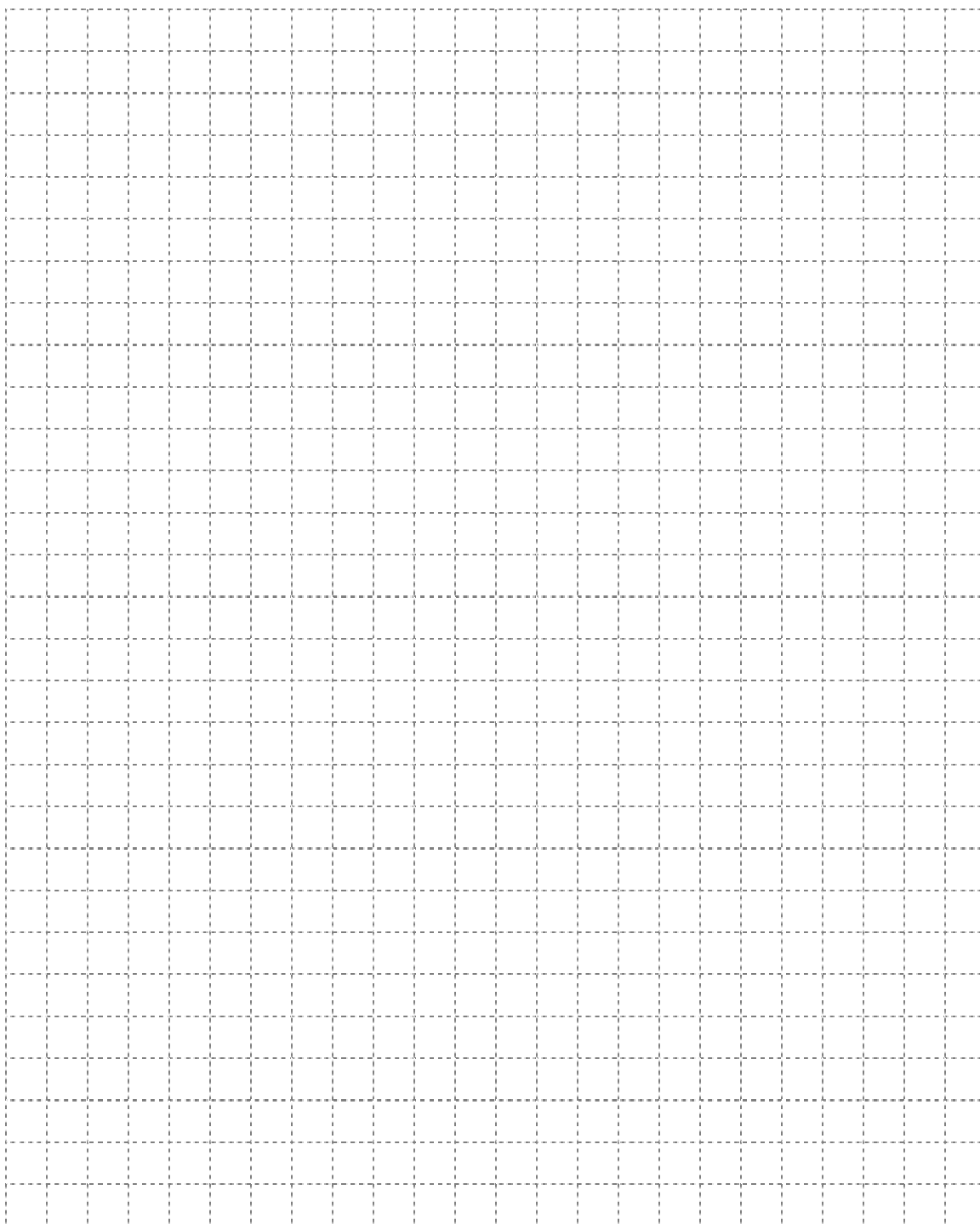
3 marks

137

Solve:

$$9^{2x+1} = 27^x$$

No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.